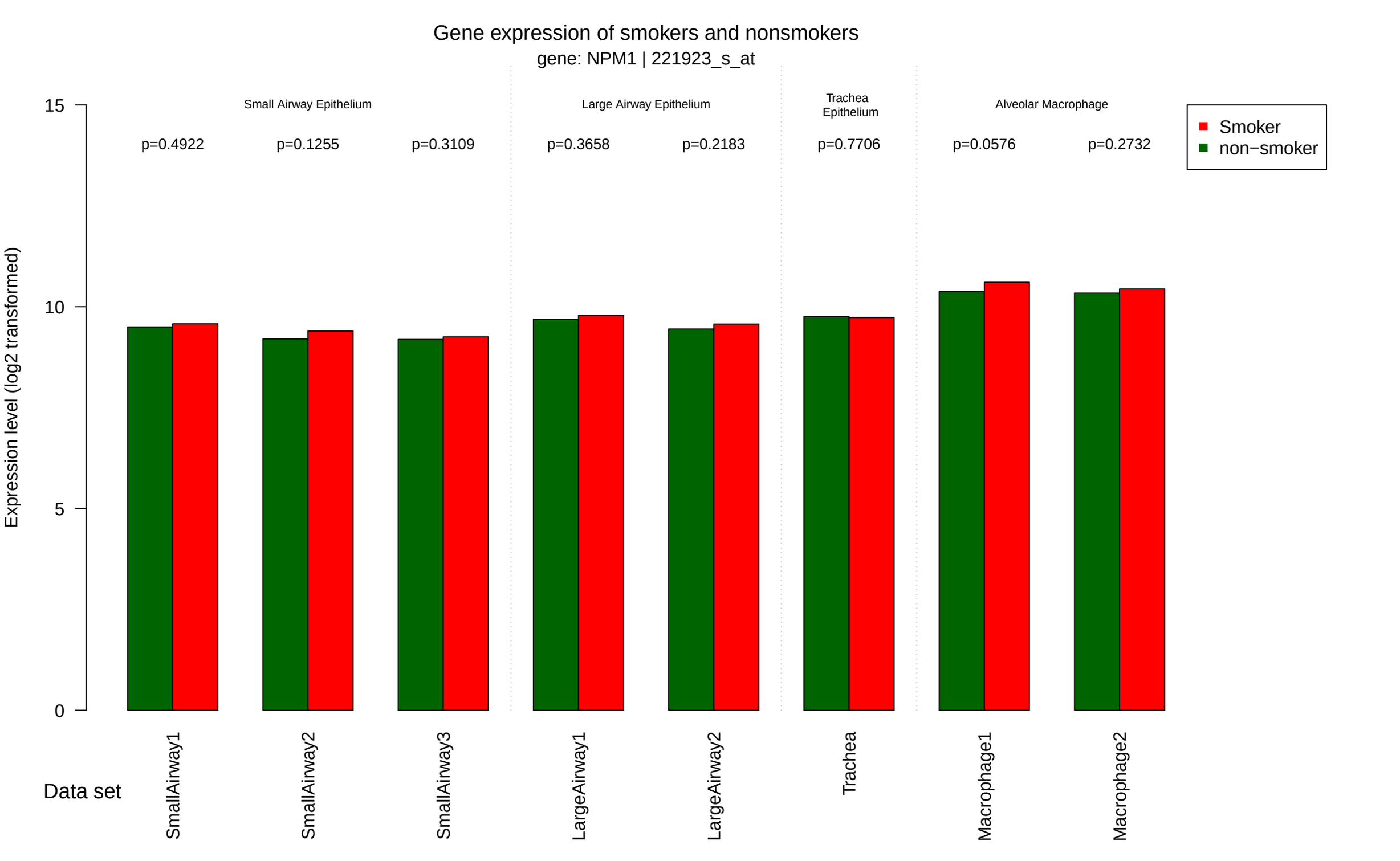
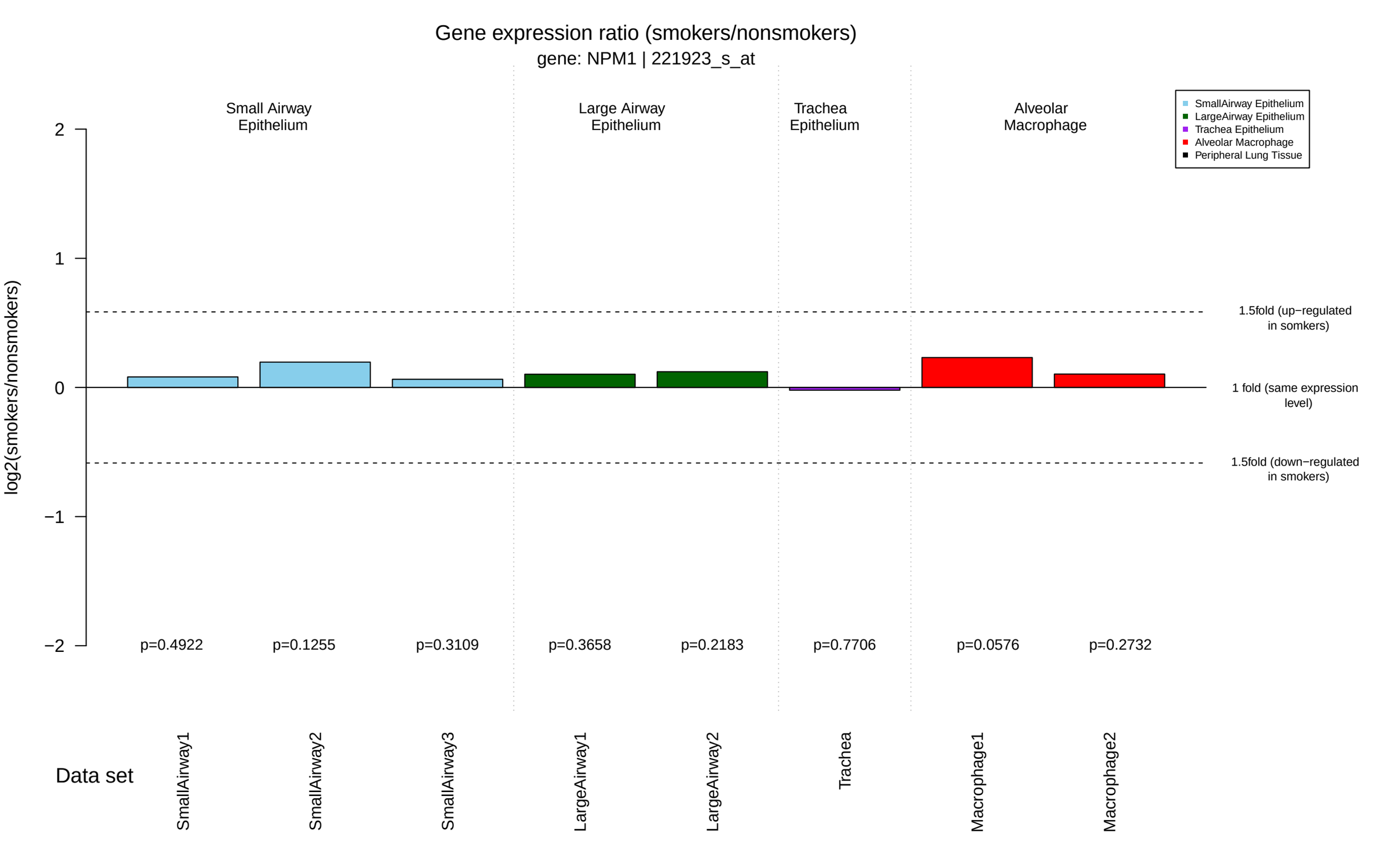
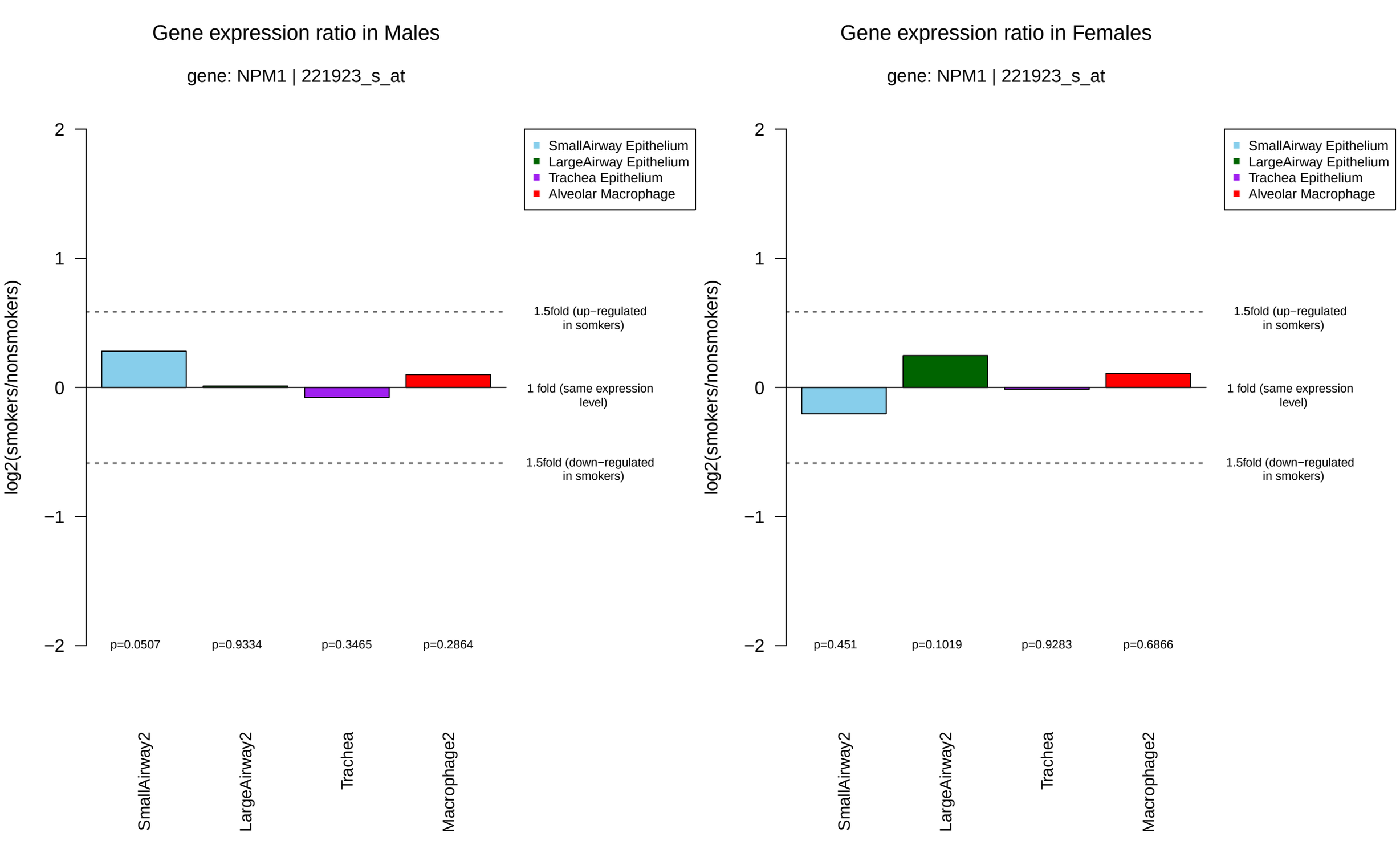
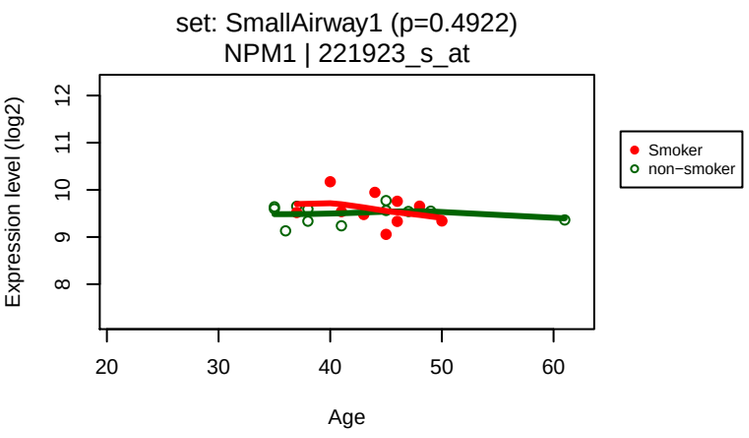
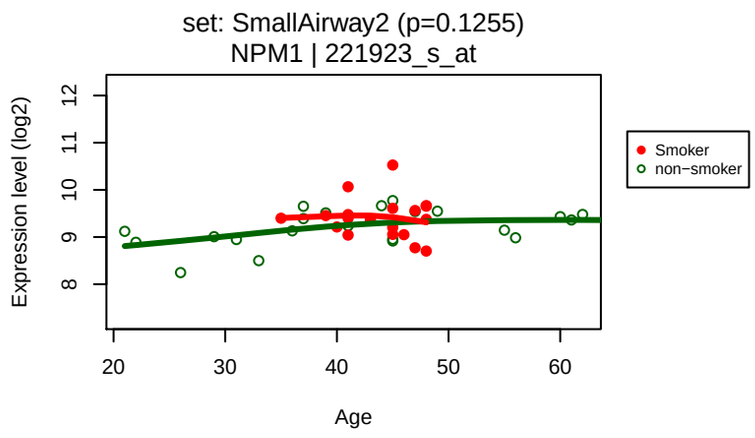
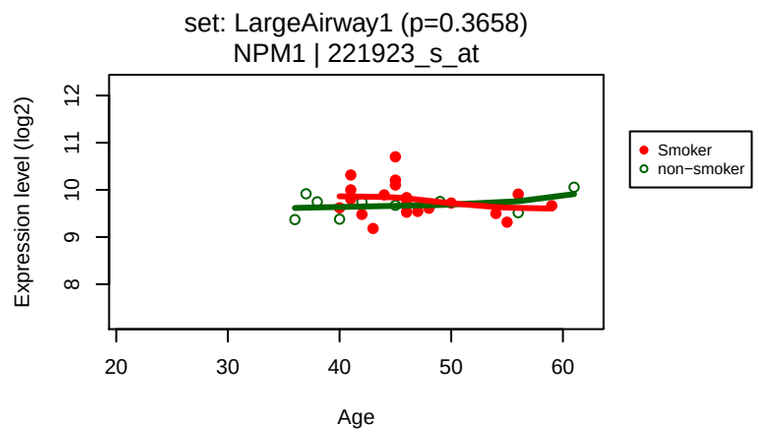

Gene expression of smokers and nonsmokers
gene: NPM1 | 221923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4922
p=0.1255
p=0.3109
p=0.3658
p=0.2183
p=0.7706
p=0.0576
p=0.2732
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

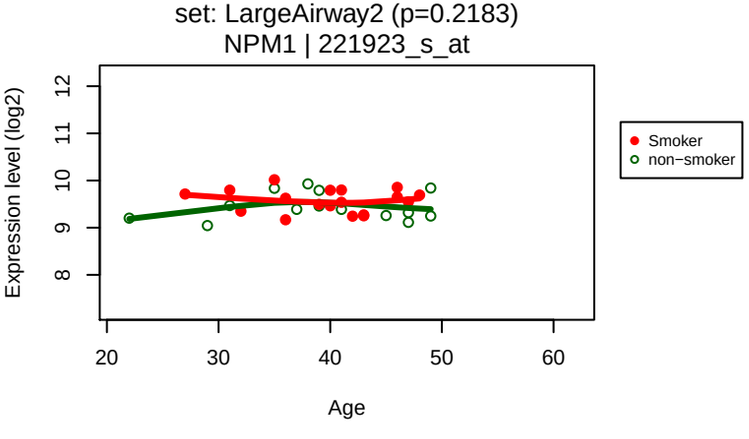

Gene expression ratio (smokers/nonsmokers)
gene: NPM1 | 221923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4922
p=0.1255
p=0.3109
p=0.3658
p=0.2183
p=0.7706
p=0.0576
p=0.2732
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

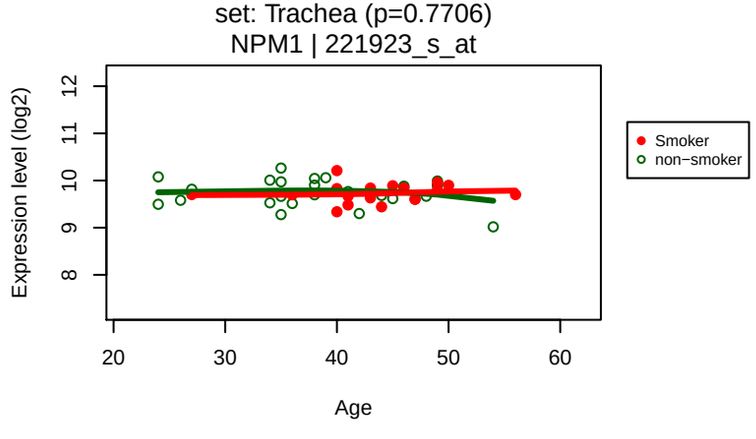

Gene expression ratio in Males
gene: NPM1 | 221923_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0507
p=0.9334
p=0.3465
p=0.2864
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPM1 | 221923_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.451
p=0.1019
p=0.9283
p=0.6866
SmallAirway2
LargeAirway2
Trachea
Macrophage2

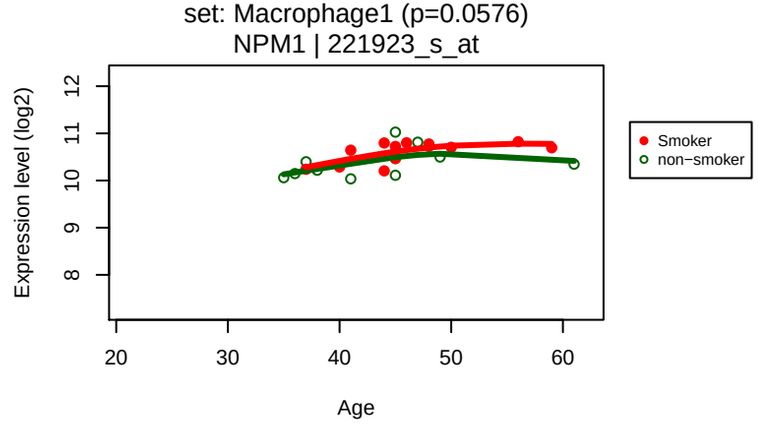

set: SmallAirway1 (p=0.4922)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker

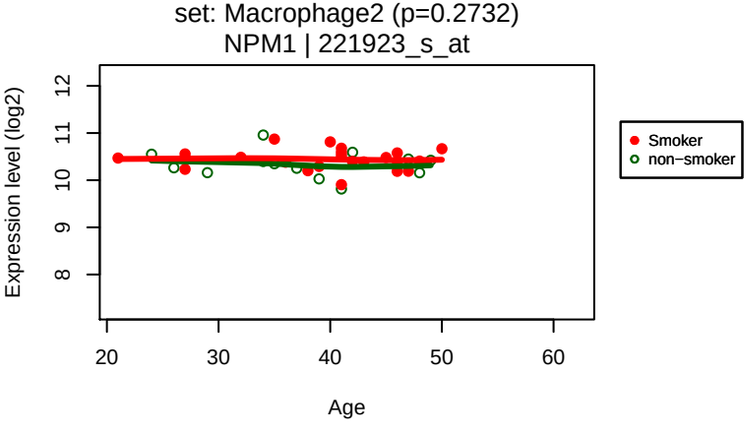

set: SmallAirway2 (p=0.1255)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker

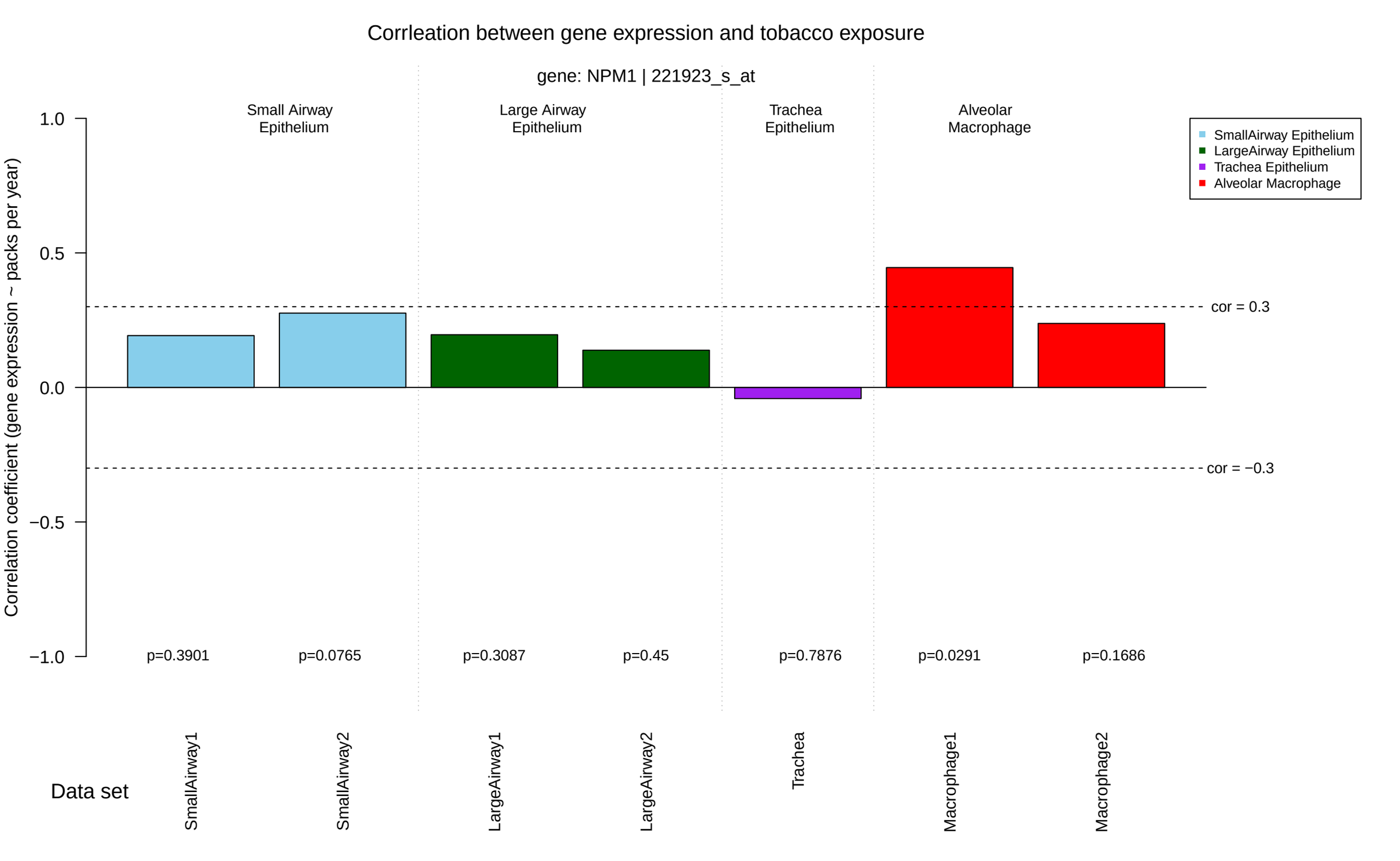

set: LargeAirway1 (p=0.3658)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2183)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7706)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0576)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2732)
NPM1 | 221923_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPM1 | 221923_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3901
p=0.0765
p=0.3087
p=0.45
p=0.7876
p=0.0291
p=0.1686
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2